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Prior elicitation for Bayesian generalised linear models with application to risk control option assessment

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ABSTRACT

A pragmatic approach to prior elicitation was developed to elicit the parameters and model structure for Bayesian generalised linear models. Predictive elicitation of subjective probability distributions was used to evaluate Risk Control Option (RCO) effectiveness for reducing the risk of ship collisions in Australia's Territorial Sea and Exclusive Economic Zone. The RCOs considered were pilotage, Vessel Traffic Services (VTS) and Ships' Routeing Systems (SRS). Predictive relationships with key covariates were documented. Distance from the Territorial Sea Baseline was important for all RCOs, and aggregate measures of shipping traffic patterns such as volume and the distribution of course over ground headings were related to the effectiveness of both VTS and SRS. A synergistic interaction between pilotage and VTS effectiveness was predicted. The elicitation method enabled a practical approach to eliciting subjective probability distributions while accounting for the complexity and myriad factors that contribute to challenging problems. The approach supports coherent updating given new information, and so can be used to support evidence based decision making.

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1. Introduction

Safety analysis is concerned with making decisions while uncertain about the relationship between possible actions and outcomes. Bayesian decision theory provides a coherent approach to decision making under uncertainty [1]. Subjective probability is the language used to capture uncertainty [2] and accommodates both aleatory and epistemic uncertainty [3]. If empirical data are available, then new information can coherently be incorporated via statistical models, that is, Bayesian learning can occur. This process ideally begins by documenting the available expert knowledge and uncertainty. The elicitation of expert opinion using subjective probability distributions rather than eliciting scores or point estimates (such as a mean) is a desirable goal when assessing quantitative risk [4], and substantial effort has been devoted to the elicitation of subjective probability distributions (as opposed to point estimates) of unknown parameters over the years [5,6].

The contribution of subjective probability distributions by experts forms a foundation for evidence based decision making. Some examples of application domains are shipping safety [7], reliability growth assessment for the development of modular software or hardware systems [8], public policy [9], estimating costs and risks for space systems [10,11], nuclear plant safety [12], hydrogeology and water treatment [13]. Well-suited domain experts often have busy schedules that are demanding of their valuable expertise and skill set. Yet complex problems often have many risk factors to consider. A successful elicitation method that allows for Bayesian updating of unknown parameters must permit non-statisticians to contribute their judgements and also mitigate the required elicitation workload to a manageable level.

For example, the assessment of risk for shipping requires the consideration of multiple risk factors such as ship characteristics, vessel traffic patterns, physical factors and the effectiveness of any deployed Risk Control Options designed to reduce risk [14]. A popular method for eliciting expert opinion in the risk analysis of shipping safety is Bayesian networks [15,16]. The expert subjective probability assessments are usually restricted to point estimates, which likely give an incomplete summary of the underlying uncertainty for a given scenario. Many elicitation targets for shipping risk would be more naturally represented by continuous distributions, which may be approximated within a discretised Bayesian network with increasingly fine partitions, but at the cost of complicating the Bayesian network structure and hence elicitation load [17]. Moreover, assessing uncertain model structure for Bayesian networks is time-consuming and difficult [18], and rarely completed. Nevertheless, despite the typical simplifications of point estimates and an assumed known model structure, the increasing use of Bayesian networks has led to recognition of the pressing need to manage the elicitation workload for shipping risk applications [16]. One alternative is to

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develop a prior for unknown coefficients in a regression function that relates risk factors to the exposure of interest. For shipping risk, one approach is suggested by Merrick and van Dorp [19], where point estimates of the relative likelihoods of collision were elicited for various scenarios that form the design points in a regression function.

The uncertainty that experts have under a given scenario may be relevant and so eliciting subjective probability distributions may be preferred to point estimates [4]. Eliciting expert subjective probability distributions has a recognised and important role in the risk analysis of shipping safety [7]. On the other hand, directly eliciting subjective probability distributions for regression coefficients is a difficult task compared to observable quantities [6,20]. A multivariate normal prior is a typical choice for linear and generalised linear models [21], but targeting a multivariate normal prior would necessitate the elicitation of means, variances and covariances. Eliciting these statistical moments is a difficult task for experts [1,6] even without the added complication of introducing known covariates into a regression model.

The assessment of linear model parameters by eliciting from experts their subjective probability distributions of predicted responses, as opposed to point estimates or directly targeting the regression parameters, began with the normal linear model proposed by Kadane et al. [22]. This approach requires the expert to consider responses conditional not only on covariates but also realisations of the response at different design points. A similar approach extends to logistic regression [23] and other generalised linear models [24]. This latter approach allows for piecewise linear models but assumes independence among parameters that correspond to different covariates.

Another predictive approach was elaborated for generalised linear models using conditional mean priors [25]. The expert in this case contributes a distribution, perhaps based on elicited moments or modes, conditional on the values of the covariates at a design point. The form of the prior may depend on the chosen link function and likelihood. However, as mentioned above, a typical choice of prior is multivariate normal independent of the choice of likelihood and link function (e.g., [24,26]). Such multivariate normal priors for generalised linear models are the focus of this paper.

A technique is proposed that elicits subjective probability distributions conditional on scenarios or design points. The proposed approach is a conditional mean prior technique, but differs from [25] in that experts are always asked to provide fractiles (equivalently quantiles or percentiles) of predictive subjective probability distributions. A more important differentiation is the form of the induced prior, which unlike [25] is multivariate normal no matter the choice of link and likelihood function. The approach allows for dependence among all unknown parameters of the linear model including continuous and categorical predictors and all possible interactions, in short, any model that can be represented as a GLM with a linear combination of basis functions. In some cases, the model structure is unknown a priori and must also be assessed from the expert elicitation, and the proposed technique is robust to this situation. The elicitation can accommodate new sources of information, whether from new elicitation or from empirical data, by Bayesian updating and so forms a foundation for evidence based decision making. An example is given for the assessment of Risk Control Options for shipping collision risk in the Territorial Sea and Exclusive Economic Zone that surrounds mainland Australia.

2. Generalised linear model

The generalised linear model (GLM) has three components [27]. The first component specifies an observation model, $p(y_i|\theta_i, \xi)$, for data y_i conditional on the expected response $\mathbb{E}[y_i] = \theta_i$ and any additional parameters ξ used by the observation model chosen from the exponential family. The second component is the *linear predictor*, η_i, that is linked to the expected value of the response. The linear predictor depends on the $p \times 1$ vector x_i of the known covariates evaluated at the design point (or "scenario"), and the $p \times 1$ vector of unknown parameters β,

$$\eta_i = x_i^\top \beta, \tag{1}$$

where the x_i may encode continuous covariates, factors, polynomials or other choice of basis functions. The third component is an invertible *link function*, $g(\theta_i) = \eta_i$, that links the expected response to the linear predictor.

In a Bayesian GLM, a normal prior is often chosen for the parameters β. A normal prior is the conjugate prior for the normal linear model [28], which is a GLM with a Gaussian observation model and identity link function. Although a conjugate prior is sometimes available, this is not the case for all choices of Bayesian GLMs. However, the posterior distribution will under some regularity conditions be approximated by a multivariate normal for large sample sizes [29]. Moreover, a normal prior includes commonly applied basis function models and Gaussian process models for the linear predictor in GLMs [21,30]. The normal prior specification can also ease comparison with other prior structures that may be contributed in the form of hierarchical models, where random effects are often modelled through a Gaussian process prior to capture spatial or temporal dependence [31] or by a multilevel normal prior in the case of exchangeable random effects [21]. A normal prior derived from expert opinion is therefore sought for the unknown coefficients β in Bayesian GLMs.

3. Conditional elicitation step

3.1. Elicitation of subjective conditional probabilities

The target of the elicitation will be the unknown θ. Typically θ is defined such that it has an direct interpretation relative to potential observables and expert knowledge. For example, in a Bernoulli response model where the y_i correspond to successes and failures, $\theta_i = \mathbb{E}[y_i]$ is interpreted as a proportion or probability of an event occurring; in a Poisson response model where the y_i correspond to counts, θ_i is interpreted as an intensity that may relate to the rate of an event occurring over a given time interval, and so on.

The expert (or experts if participating in a group elicitation) are asked to provide their subjective probability estimates for θ_i conditional on the design point x_i. This is the independent conditional mean prior approach where it is useful to assume that expert contributed responses are conditionally independent given the covariates [25]. Another approach is to ask questions about conditional distributions to elicit dependence among observations at different design points (e.g.,[22]). This approach is intractable for many forms of generalised linear models [25]. Moreover, eliciting dependencies requires additional questions, which is an important consideration given the already limited time and statistical resources available for a given elicitation or relevant domain experts (Section 1). Such approaches therefore may prompt experts by making typical proposals based on the model structure, which may run the risk of anchoring an expert to the proposed value rather than eliciting their true opinion. The selection of design points for an independent conditional mean prior elicitation is an experimental design question discussed further in Section 4.2.

A method that elicits fractiles is the preferred approach over other approaches that ask for moments [5,6]. Experts have difficulty assessing how sensitive the mean of a distribution is to its tails, whereas the probability statements encoded by fractiles are easier to elicit [1]. We adopt a normal model for $\eta_i|x_i$ with the conditional mean and variance formed into the parameter vector $\phi_i = [m_i, v_i]^\top$. In the generalised linear model framework, the vectorised link function $g(\cdot)$ is monotonic and preserves fractiles. Therefore, conditional on a design point x_i, an elicitation of fractiles for the target $\theta_i = g^{-1}(\eta_i)$ directly translates into fractiles for the conditional normal distribution of the linear predictor, $g(\theta_i|\eta_i) = \eta_i|x_i \sim N(m_i, v_i)$. The parameters ϕ_i are chosen to minimise the information lost by approximating the elicited fractiles by a normal distribution as follows.

Assume that the expert (or group of experts) have given F_e fractiles that describe the target conditional subjective probability distribution, $P_e(\theta_i|x_i)$, noting that θ_i depends on the known x_i. Given the monotonic link function $g(\cdot)$, corresponding fractiles are imposed for the conditional probability distribution $P_e(\eta_i|x_i)$. Assemble these latter fractiles f_j for the linear predictor in increasing order with $j \in \{1, \ldots, F_e\}$. Let f_0 and f_{F_e+1} be determined by the lower and upper bounds (possibly infinite) of the support of θ_i and form the vector, $f = [f_0, f_1, \ldots, f_{F_e}, f_{F_e+1}]^\top$. Let the probability intervals determined by these elicited fractiles be denoted by l_j, $j \in \{1, \ldots, F_e + 1\}$. For example, if the quartiles and median are elicited, then $l_j = 0.25$ for $j = 1, \ldots, 4$.

Now let the conditional subjective probability distribution function parametrised by ϕ_i be denoted by $P_s(\eta_i|\phi_i)$. The fitted probability intervals are obtained by,

$$z_j = \int_{f_{j-1}}^{f_j} dP_s(\eta_i|\phi_i) = P_s(f_j|\phi_i) - P_s(f_{j-1}|\phi_i),\ j = 1, \ldots, F_e + 1.$$

The Kullback–Leibler divergence of $P_{s|i}$ from $P_{e|i}$, where subscripts denote dependence on ϕ_i and x_i respectively, provides the expected loss for reporting $P_{s|i}$ when the true model is $P_{e|i}$ given a logarithmic scoring function [29]. The Kullback–Leibler divergence is approximated by,

$$\mathrm{KL}(P_{e|i}||P_{s|i}) = \int \log \frac{dP_{e|i}}{dP_{s|i}} dP_{e|i} \approx \sum_j \log\left(\frac{l_j}{z_j}\right) l_j,$$

where $P_{e|i}$ is absolutely continuous with $P_{s|i}$. The discretisation into probability intervals l_j and z_j results in a loss of information [32]. The discretised approximation converges to the divergence measure with increasingly fine partitions [33], that is, increasing F_e.

The parameters ϕ_i are chosen so as to minimise the approximate KL divergence of $P_{s|i}$ from $P_{e|i}$,

$$\hat{\phi}_i = \arg\min_{\phi_i} \sum_j \log\left(\frac{l_j}{z_j}\right) l_j.$$

The predicted fractiles that correspond to the fitted subjective conditional probability distribution $P_{s|i}(\eta_i|\hat{\phi}_i)$ impose predicted fractiles for the target subjective conditional probability distribution, $P_{s|i}(\theta_i|\hat{\phi}_i)$. The expert may also assess the shape of the subjective probability density function of the target, $p_{s|i}(\theta_i|\hat{\phi}_i) = dP_{s|i}(\theta_i|\hat{\phi}_i)/d\theta_i$, which is more interpretable than a cumulative probability density function [34]. An iterative process, described in detail below, continues until the experts are satisfied that the distribution for θ_i induced by $P_{s|i}(\eta_i|\hat{\phi}_i)$ is a reasonable model of their beliefs.

3.2. Elicitation steps for experts

Before undergoing an elicitation session, experts were first educated about subjective probability, trained on heuristics and biases commonly encountered in expert elicitation exercises, and presented practice examples before beginning the elicitation [5,6,35]. The elicitation process for a given scenario was iterative and begins with an initial assessment for the fractiles f_j, $j \in \{1, \ldots, F_e\}$. A number of alternatives for eliciting fractiles are available. One recommended approach is the elicitation of quartiles, which elicits the lower quartile ($f_{1/4}$), median ($f_{1/2}$) and upper quartile ($f_{3/4}$). This method was used by Garthwaite [36] in a regression setting and is one of the more common approaches to eliciting fractiles [5]. For a given design point defined by an elicitation scenario x_i, the following questions pose the elicitation such that the experts perform judgements of equal odds:

1. What value do you believe gives a 50% chance that the true answer is lower? (This is $f_{1/2}$).
2. Assume that the true value is really **below** $f_{1/2}$. Given this information, what value do you believe gives a 50% chance of being above or below the true value? (This is $f_{1/4}$).
3. Assume that the true value is really **above** $f_{1/2}$. Given this information, what value do you believe gives a 50% chance of being above or below the true value? (This is $f_{3/4}$).

For each scenario, the following steps are then performed by the experts:

1. Elicit initial fractile assessments by the quartile method (vertical black dashed lines in Fig. 1).
2. Plot the corresponding fitted fractiles and density curve from the fitted density $p_{s|i}(\theta_i|\hat{\phi}_i)$ (middle three vertical blue dashed lines and blue curve in Fig. 1). If the values for the matching fractiles are not acceptable than return to Step 1 and adjust as necessary.
3. Predict new fitted fractiles from $p_{s|i}(\theta_i|\hat{\phi}_i)$. The extreme deciles are presented by the leftmost and rightmost vertical dashed blue lines in Fig. 1. If the values for the prediction fractiles are not acceptable than return to Step 1 and adjust as necessary.
4. Consider any other relevant fractiles or cumulative probabilities from the fitted distribution. If these values are not acceptable than return to Step 1 and adjust as necessary.
5. Accept the elicited subjective probability distribution $p_{s|i}(\theta_i|\hat{\phi}_i)$ as a reasonable assessment of the expert opinion.

In Step 1, eliciting more fractiles than free parameters is termed "overfitting" [6], which is a desirable feature in this context. Overfitting averages uncertainty in the parametric representation of the expert opinion across multiple observations and encourages the elicitation process to directly confront the limitations and advantages of the parametric model representation. Feedback is provided in Steps 2 through 4. In Step 2, presenting the fitted fractiles after eliciting the quartiles in Step 1 helps limit the possibility of experts anchoring on suggested values [6]. Also in Step 2, the fitted probability density function is presented for graphical feedback. Further feedback is provided by presenting the values of the fractiles in Steps 3 and 4. This approach to elicitation helps acknowledge limitation in the parametric representation of expert judgement while seeking a reasonable interpretation of uncertainty.

4. Prior estimation step

4.1. Prior distribution of β

The expert elicitations induced for the linear predictor are each normally distributed, such that,

$$\eta_i|x_i \sim N(m_i, v_i), \quad i = 1, \ldots, n. \tag{2}$$

The m_i and v_i are conditionally independent from other elicitations given the design point x_i. The n design points are assembled into the $n \times p$ design matrix X with rows given by $x_i^\top$ for $i = 1, \ldots, n$ that is assumed to have full column rank $p \le n$. The distribution of η conditional on the elicited data is then proportional to the exponential of a quadratic form in β (Appendix A),

$$\prod_{i=1}^{n} p(\eta_i|m_i, v_i) \propto \exp\left\{-\frac{1}{2}(X\beta - m)^\top V^{-1}(X\beta - m)\right\}, \tag{3}$$

where $m = [m_1, m_2, \ldots, m_n]^\top$ and $V = \mathrm{diag}(v_1, v_2, \ldots, v_n)$ is a diagonal covariance matrix. The distribution of the unknown parameter vector β is then multivariate normal with mean and covariance given by (Appendix A),

$$\mu = \left(X^\top V^{-1} X\right)^{-1} X^\top V^{-1} m, \tag{4}$$

$$\Sigma = \left(X^\top V^{-1} X\right)^{-1}. \tag{5}$$

The expressions of Eqs. (4) and (5) are equivalent to the weighted least squares estimate in a regression with heterogeneous variance [37]. The induced prior is $p(\beta) = N(\beta|\mu, \Sigma)$.

4.2. Elicitation design

A design is the selection of n design points assembled into the $n \times p$ design matrix X with rows given by $x_i^\top$ for $i = 1, \ldots, n$. Every Bayesian

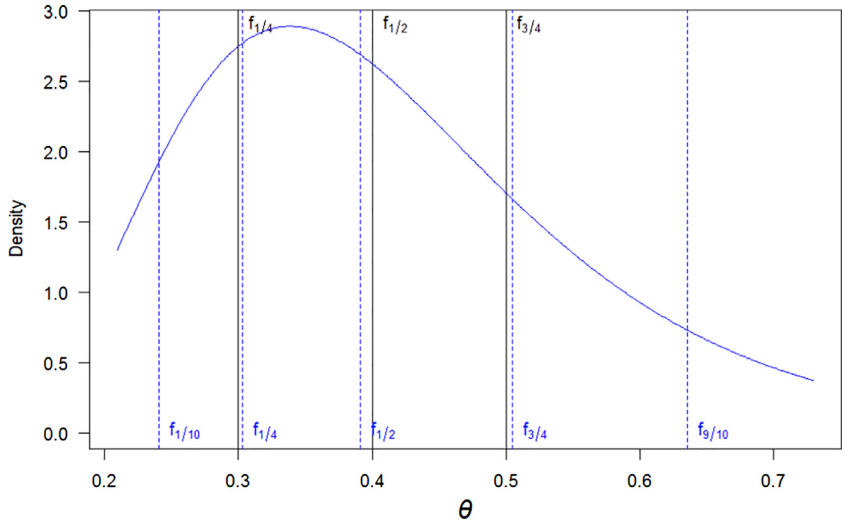


Fig. 1. Example of graphical feedback to experts for an elicited distribution. The elicited target is θ. Probability density function (blue curve) with x-axis bounded by the 90% central CI of the fitted distribution. The elicited fractiles are displayed as the black solid vertical lines, and the fitted fractiles as the blue dashed vertical lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

GLM with a normal prior has a corresponding design for which the independence assumption in a conditional mean prior approach is appropriate [25]. Note, for a prior developed from such a design, predictions at new design points (see Section S1 of the Online Supplementary Material (OSM)) will generally not be independent. The focus then is to choose design points that will help approximate the conditional independence assumption before conducting an elicitation session. Choosing such a design can be difficult because it depends on expert opinion that is the focus of the elicitation.

The choice of a well-separated design lessens dependence in the conditional means [25]. A well-separated design is one that ensures design points are not clustered close together in the covariate space. Informally, this is achieved by spreading out observations in the covariate space. Broad guidance on how to achieve a well-separated design can be found in the statistical design literature. In particular, with the normal prior of Section 2, the literature for optimal design of linear models could be applied. See for example [38], or [39] for a review of Bayesian optimal design. The optimal design literature also provides guidance on the number of elicitations (sample size) needed, which will tend to increase as the complexity of the model increases. Minimally, at least $n \geq p$ elicitations will be required for p unknown parameters to fulfil the requirement that the design matrix X is full column rank.

4.3. Diagnostics for the design and elicited prior

The problem of dependence in the conditional means is equivalent to the well-recognised issue of collinearity in regression problems, and so a wide range of regression diagnostics can then be applied [25]. One such measure considered by Bedrick et al. [25] is the condition number of the scaled design matrix X_s, with each column given by the corresponding column in X scaled to unit length. It is related to the sensitivity of the inverse of Σ from Eq. (5) given a perturbation of X. Twice the condition number places an upper bound on the elasticity of a least squares estimate, as mediated by the inverse with respect to a change in X_s; a broad guideline is that a condition number for X_s in the 15–30 range may correspond to dependency in the design matrix [40]. This diagnostic can be applied to assess possible issues that could arise from a choice of design.

The above diagnostic considers only the choice of design X. Once the elicitations are collected, the condition number can similarly provide information about the sensitivity of the prior to changes in either X or the elicited V. An analogous scaling would be the $n \times p$ matrix L_s that is composed of the scaled columns of L where $L^\top L = \Sigma^{-1}$. If regression diagnostics suggest that collinearity is an issue, then for a given model either new data can be collected or additional prior information can be introduced [40]. If the model structure is uncertain, then a new model could be considered with the correlated predictors removed. The question of model structure uncertainty is discussed further in the next section.

4.4. Model structure uncertainty

Many indirect elicitation techniques are predicated on knowledge of the model structure during the course of the elicitation session [22–24]. Ideally such a model structure is itself constructed from prior expert opinion. The prior estimation can be done as the elicitation progresses as soon as more than p scenarios are elicited. This allows prior feedback to the expert during the elicitation [*sensu* 22, 24].

However, in principle, providing such convenient hints could lead to anchoring and may provide undue influence if the functional form is uncertain. Some covariates initially thought important, and which may advise the choice of design (Section 4.2), may over the course of an elicitation session prove less so or may be replaced by a more relevant choice. Conditional on a choice of predictors, different basis functions could be applied as more design points are considered. For example, a linear fit could be subsequently compared to a quadratic relationship. If the functional form is uncertain, then the elicitation may be in part conducted to investigate the model structure. In that case finalising the model structure before the elicitation would be counter-productive. For these reasons, it may be useful to separate the data collection from the data analysis.

In the analysis stage that follows an elicitation, different choices of covariates will lead to modified versions of X and θ that depend on the choice of model M_j. An asymptotic approximation to the log Bayes factor of two alternative models is given by minus twice the Schwarz criterion [41,42], which is commonly referred to as the Bayesian Information Criterion (BIC). Analogous to a linear regression, using Eq. (3) this approximation is given by,

$$\begin{aligned} \mathrm{BIC}_j &= -2\log p\left(\hat{\beta}\middle| X, m, V, M_j\right) + p_j \log n \\ &\propto \left(X\hat{\beta} - m\right)^\top V^{-1}\left(X\hat{\beta} - m\right) + p_j \log n, \end{aligned} \tag{6}$$

where the BIC is evaluated by substituting $\hat{\beta}|M_j = \mu|M_j$ into Eq. (6) and setting p_j to the dimension of the vector $\beta|M_j$. Lower values of BIC correspond to better fitting models.

5. Illustrative example

5.1. Expert judgement and shipping collision risk

Assessing the amount of risk reduction that may be achieved by the implementation of a risk control option (RCO, also known as a risk control measure or risk intervention measure) is an important factor for assessing shipping safety [43]. Expert elicitation is an important knowledge source because of the sparse data available for low-probability events such as collisions and how risk may be reduced by various RCOs. Discrete Bayesian networks constructed from expert judgement have become very popular for this purpose [16] with applications to shipping collisions for liquefied natural gas carriers [44] and oil transportation [17], for example.

An alternative approach to incorporating expert judgement into shipping collision risk was applied by Merrick et al. [45] to oil transportation in Prince William Sound. In that study, scenarios were described by covariates that quantified features as visibility and ship traffic information. Experts contributed a point estimate of the ratio of probability of a collision or other type of accident for a scenario with one covariate value modified compared to a baseline scenario. The relationship between relative accident probabilities and the covariates were then estimated by loglinear regression. The baseline probability of accident must be obtained separately to estimate the absolute accident probabilities [45]. This approach was extended by Merrick and van Dorp [19], Szwed et al. [46] to a fully Bayesian analysis for estimating the risk of collision for ferry service in the United States.

The approach applied here, as described in Sections 2–4, differs from the foregoing studies of shipping risk in that subjective probability distributions are elicited from experts. As noted above, subjective probability distributions are a natural way to accommodate expert estimates of risk reduction through risk control options in shipping applications [7,17]. The regression approach of Sections 2–4 could be applied to estimate either relative collision probabilities, as in [19,45,46], or absolute collision probabilities. The following provides an example application of estimating and predicting the effectiveness of risk control options for shipping collisions under alternative scenarios.

5.2. Problem formulation

The Australian Maritime Safety Authority (AMSA) sought to predict the effectiveness of risk control options (RCOs) at a national scale for long term strategic planning purposes. In this project, the effectiveness of RCOs was evaluated for the risk exposure endpoints (REEs) of collisions, allisions and powered groundings; only the first REE (collision risk) is considered here. The RCOs considered were pilotage, Vessel Traffic Services (VTS) and Ships' Routeing Systems (SRS). In question was the amount of risk reduction achievable by implementing these RCOs either individually or in combination. Empirical information (incident data) on the annual rate of these REEs in Australia, and hence the risk-reduction benefits provided by RCOs, is limited. AMSA therefore sought opinion from experts who were familiar with Australian shipping to quantify their risk reduction effect. The expert elicitation process assessed uncertainty and analysed the relationships among the identified REEs, RCOs and other risk generating factors. The elicitation session used behavioural aggregation [5,47] in a group setting to allow the experts to discuss the effectiveness of RCO across a variety of scenarios. Five AMSA experts representing different domains of RCO expertise, experience and background cooperatively participated in the elicitation session.

5.3. Defining the effectiveness of risk control options

The Risk Exposure Endpoint of collisions was defined as contact between two ships that are both under power, which results in damage. The experts noted the possible interactions among RCOs. The effectiveness of the RCOs were considered individually and in combination. The definitions of the Risk Control Options (RCOs) were as follows:

Pilotage: A service provided by an experienced mariner who has expert knowledge of local conditions and ship handling and who is generally licensed by a maritime authority. A pilot is an external adviser, engaged to assist shipmasters in the navigation of vessels in confined or environmentally sensitive waters.

Vessel Traffic Services (VTS): A service implemented by a Competent Authority, designed to improve the safety and efficiency of vessel traffic and to protect the environment. The service should have the capability to interact with the traffic and to respond to traffic situations developing in the VTS area. There are currently three types of services that can be provided by a VTS: Information Service, Traffic Organisation Service, Navigational Assistance Service.

Ships' Routeing Systems (SRS): Any system of one or more routes or routeing measures aimed at reducing the risk of casualties. Examples are Traffic Separation Schemes, two-way routes and areas to be avoided (either mandatory or recommended).

The effectiveness of Risk Control Options was assessed by targeting the rate ratio of a square nautical mile "scenario location" given a reasonable deployment of a RCO or combination of RCOs in the scenario neighbourhood, which was defined as the square with dimensions that extended 10 nautical miles from the scenario location in the cardinal directions. The rate ratio is defined by the annual number of collisions in the scenario location that would be expected given a RCO implementation in the scenario neighbourhood divided by the number of collisions that would be expected in the absence of *any* of the RCOs (including those RCOs currently implemented). The rate ratio was thus defined as,

$$\theta = \frac{\text{Number of collisions with RCO(s)}}{\text{Number of collisions without any RCOs}},$$

considered per year at the scenario location.

The rate ratio gives a quantitative measure of RCO effectiveness. For example, if introducing pilotage reduces the expected number of collisions to 10% of the number that would be expected for the same scenario without any RCOs, then the rate ratio is 0.10; if the expected number of collisions with pilotage is 30% of that expected without any RCOs then the rate ratio is 0.30, and so on. A scenario location with a rate ratio of one means that a RCO has no effect, that is, the incident rate is the same both with and without the RCO. A scenario location with a rate ratio greater than one means that implementing the RCO actually increased the incident rate. A scenario location with a rate ratio of zero means that the RCO eliminated all risk for the given scenario. The experts were never asked to assess the rates of collision themselves, which can be difficult for such rare events; instead, experts were only asked how the RCO implementation might change the incident rate.

The adjective "reasonable" was included in the above RCO definition to account for the practicalities of deployment in the scenario neighbourhoods. For example, pilots were assumed to be trained and qualified. On the other hand, compliance for a single RCO may not be reasonably expected to be observed by all craft in all circumstances, for example, a recommended SRS routeing measure in the absence of oversight by a pilot or VTS. The spatial scope considered by the elicitation workshops included the Territorial Sea and EEZ contiguous to mainland Australia.

5.4. Model setup

The number of collisions c_i in a given area a_i over a year interval was assumed to be described by an inhomogeneous Poisson point process,

$$c_i \sim Poisson(a_i \lambda_i \theta_i), \tag{7}$$

where for locations $i = 1, \ldots, n$, the area is one square nautical mile (a_i = 1 nm^2), λ_i is the baseline collision rate in the absence any risk control options, and θ_i is the rate ratio. The expected rate of collision per square nautical mile for scenario i is described by,

$$E[c_i] = \lambda_i \theta_i = \exp\left(z_i^\top \alpha\right) \times \exp\left(x_i^\top \beta\right) \tag{8}$$

where the $q \times 1$ covariate vector z_i describes information relevant to the baseline rate of collision in the absence of RCO deployment, and the $p \times 1$ covariate vector x_i describes information relevant to RCO effectiveness. Covariates may or may not be shared among z_i and x_i. The $q \times 1$ parameter vector α describes the relationship between z_i and the baseline collision rate λ_i, and the $p \times 1$ parameter vector β describes the relationship between x_i and RCO effectiveness.

The canonical log link is assumed such that,

$$\log(\lambda_i \theta_i) = z_i^\top \alpha + x_i^\top \beta, \tag{9}$$

and a multivariate normal prior is assumed for the unknown coefficients,

$$\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \sim N\left(\begin{bmatrix} \gamma \\ \mu \end{bmatrix}, \begin{bmatrix} K & 0 \\ 0 & \Sigma \end{bmatrix} \right), \tag{10}$$

where the baseline rate of collisions λ_i is considered independently of the rate ratio θ_i. The question of baseline rate of collisions is considered in an independent work [14]. Further details for this model including the choice of link function are given in Section S2 of the OSM. All elicitation questions for RCO effectiveness targeted only the rate ratio, θ_i: the target of the indirect elicitation exercise.

5.5. Covariates

Key covariates were identified by the experts at workshops held prior to the elicitation session. The RCO elicitation was focussed on SOLAS commercial vessels, but recognised that non-SOLAS vessels may influence RCO effectiveness in some areas. Summary information about vessel traffic patterns for various ship categories in the Territorial Sea and EEZ that surrounds mainland Australia were considered at the elicitation session (Section S3 of the Online Supplementary Material, OSM). These summaries and other metrics of ship traffic patterns were based on Automatic Identification System (AIS) data (see [48] for discussion of the potential uses and limitations of AIS data) for the 2014 calendar year provided by AMSA. The details of the AIS data processing and covariate construction are given in Sections S3 and S4 of the OSM.

The number of nautical miles travelled (nm; Fig. 2(a)) was assumed to have a logarithmic relationship with RCO effectiveness and was considered as a covariate at the scale of both the scenario location and the scenario neighbourhood. A $\log(nm + 1)$ transform was used for the analysis. In Fig. 2(a), corridors of varying ship traffic intensity are clearly visible at the national scale. Graphical summaries of nautical miles travelled in the scenario neighbourhood and its surrounding region were also provided to the experts (Fig. 3). The relative value of traffic density at the scenario location and scenario neighbourhood were also provided (Fig. 4, left and centre),

A similar summary of the distance to the Territorial Sea baseline to the scenario location was additionally presented (Fig. 4, right). This latter covariate, considered on the logarithmic scale for the analysis, could be important for the legal reasons affecting RCO deployment as discussed above and also could indicate proximity to the Lowest Astronomical Tide, and hence serve a proxy for the availability of sea room.

Graphical summaries of vessel course over ground headings were also provided at the scale of the scenario location and the scenario neighbourhood (Fig. 5). Vessel headings were summarised for analysis into two covariates. First, the angular mean resultant length R is a measure of the distribution of angles [49]. If course over ground for every vessel is in the same direction, then $R = 1$. By inspecting Fig. 2(a) and (b), such high values tend to appear in areas of relatively low traffic (the pink areas in the centre panel fall in the green areas of the top panel). If vessels travel in opposite direction with equal frequency then $R \approx 0$, as can be

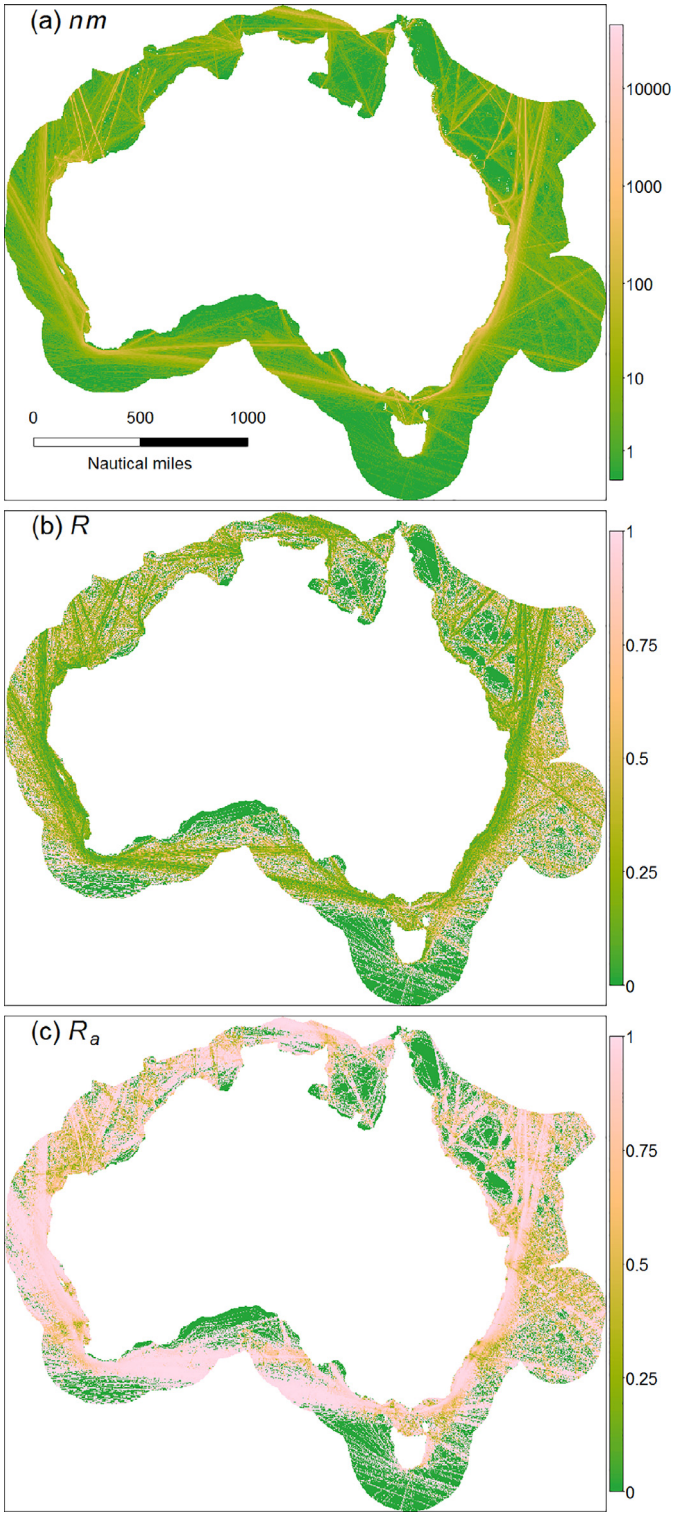


Fig. 2. Summaries of covariates for all ship categories at the square nautical mile resolution for the Exclusive Economic Zone and Territorial Sea contiguous to mainland Australia. (a) Nautical miles travelled, nm. (b) Mean resultant length, R, a measure of relative unidirectionality in vessel course over ground headings. (c) Axial mean resultant length, R_a, a measure of parallel course over ground headings. Similar summaries were provided for subcategories of ship including Class A ships with known length greater than 24 m, which contributes the majority of the AIS fixes within the spatial scope.

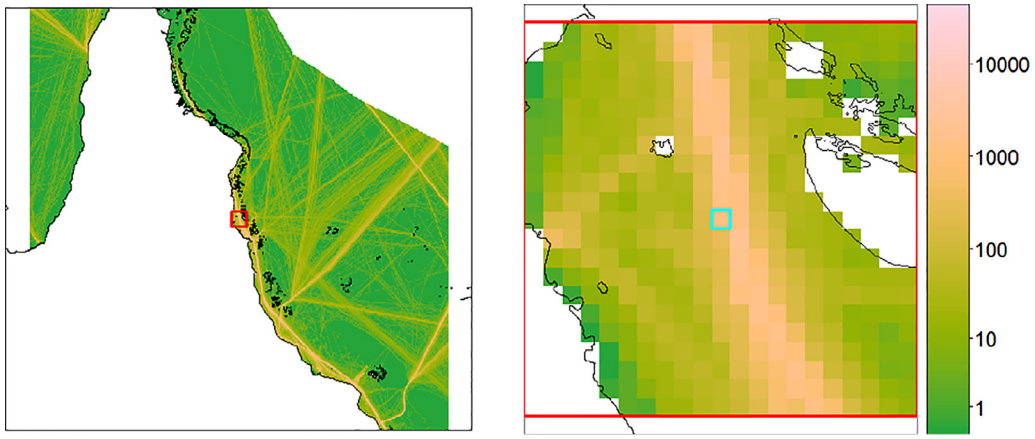


Fig. 3. Left: Nautical miles travelled at the square nautical mile resolution for the Exclusive Economic Zone (EEZ) in the 300 by 300 nautical mile area surrounding an example scenario location. The black line is the Territorial Sea Baseline. The scenario neighbourhood, extending 10 nautical miles from the scenario location in the cardinal directions, is defined by the red box, which is zoomed into on right: the scenario neighbourhood defines where the Risk Control Option is applied. The effect of the RCO on the rate ratio for collision risk is assessed in the square nautical mile scenario location (cyan box). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

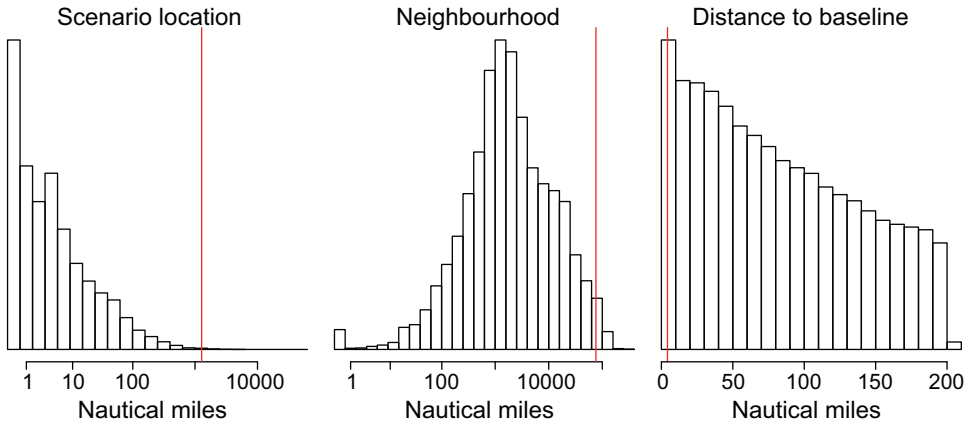


Fig. 4. Left: Histogram of nautical miles travelled in the EEZ aggregated at the nautical mile resolution. The red vertical line shows the value for the current scenario location. Centre: Nautical miles travelled aggregated over the scenario neighbourhoods that correspond to all possible scenario locations. The vertical line shows the value for the current scenario neighbourhood. Right: Distance from Territorial Sea Baseline in nautical miles for all scenario locations. The vertical line shows the value for the current scenario location. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

seen in the ship corridors located off the eastern coast of Australia in Fig. 2(b). This could occur along defined traffic corridors, for instance, which is materially different from a situation where vessels are heading every which way (where $R \approx 0$).

Second, to discriminate corridor traffic that travels in parallel opposing directions, the axial distribution was also considered [49]. The axial mean resultant length is $R_a = 1$ when every vessel is going the same direction, as is also the case for R. If equal numbers of vessels travel in opposite direction then $R_a = 1$ again, whereas $R = 0$ as mentioned above; this appears to be the case in the corridors of the eastern coast of Australia (Fig. 2). In the case where there is an orthogonal (90°) traffic crossing of two routes with identical traffic, then both $R_a = 0$ and $R = 0$. This axial alignment covariate was considered both at the scenario location and scenario neighbourhood, where it served as an indication of a defined traffic corridor.

A large amount of background knowledge was associated with each geographical location. For example, nautical charts and expert experience contained distinct ancillary information for each location. The consideration of new locations was costly to evaluate. A total of 44 elicitation scenarios were considered across a dozen geographic locations. Each of the three RCOs was considered at every location. For each scenario, graphical feedback and numeric values for the key covariates were presented. The experts also made use of the information provided by the geographic location of the scenario and the relevant information captured by navigational charts.

5.6. Results

The parameter estimates of the best fitting model by BIC (Eq. (6)) are given in Table 1. The condition number for the scaled design matrix X_s is 7.4, and the condition number for L_s, which incorporates the elicited data, is 9.3 (see Section 4.3). The independence conditional mean prior assumption appears satisfactory. Table 1 show that the pilotage and VTS RCOs have a strong synergistic effect such that the overall effectiveness of this combination was predicted to be greater than the sum of their individual effects ($P \times V$ in Table 1 shows a superadditive interaction).

The estimated effectiveness of pilotage decreased at distances far from the Territorial Sea baseline (Fig. 6, top left) with a high degree of certainty (Fig. 6, bottom left). Note the common scales for the y-axes in Fig. 6, which obscures some of the spatial variation for a given RCO (the common scale is chosen here to visualise the differences in RCO effectiveness). The effectiveness of pilotage is strongest near the territorial sea baseline. The expert commentary suggested that this might occur because of decreased relevance of pilots' local knowledge far from shore.

The estimated effectiveness of VTS also decreased at distances far from the Territorial Sea Baseline (Fig. 6, top centre). This might occur because VTS cannot deploy Traffic Organisation Services beyond the limits of the Territorial Sea, which extends 12 nautical miles beyond the Territorial Sea Baseline, under the *International Convention for the Safety of Life at Sea, 1974* (Chapter 5 "Safety of Navigation", Regulation

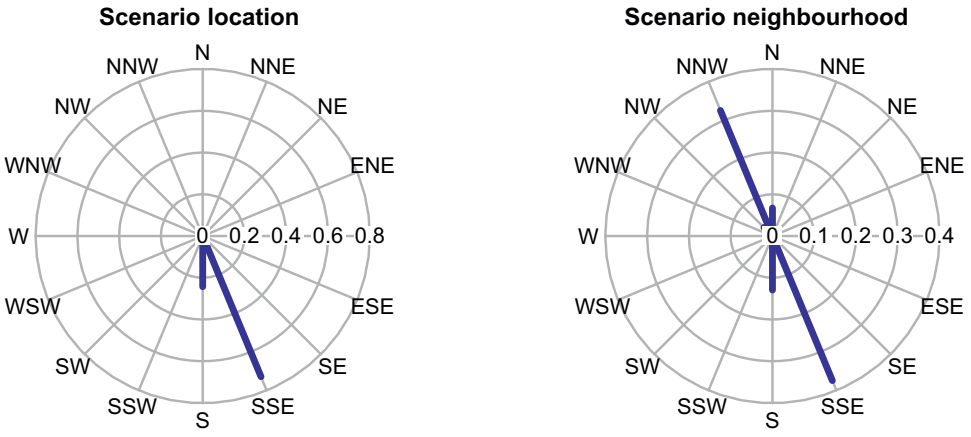


Fig. 5. Polar plots giving the proportion of course over ground headings aggregated to a 16 point compass rose at the scenario location (left) and scenario neighbourhood (right). Within the scenario location, vessel course over ground is relatively aligned primarily south southeast ($R = 0.95$, $R_a = 0.99$). Within the larger scenario neighbourhood, course headings are split north northwest and south southeast ($R = 0.10$, $R_a = 0.87$).

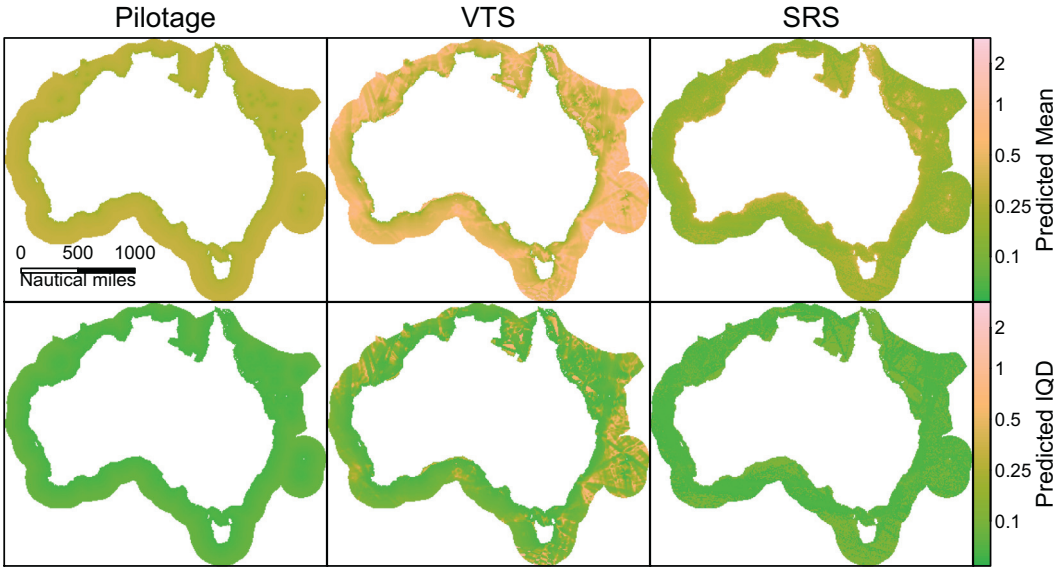


Fig. 6. Predicted rate ratio for RCOs taken individually (no interactions) on the collision REE. *Top:* Predicted mean of rate ratio. *Bottom:* Predicted interquartile distance (IQD). Note common y-axes for all RCOs.

12, Provision 3). The expert commentary suggested that Traffic Organisation Services are a key component of VTS effectiveness. Outside of the Territorial Sea, the uncertainty of VTS effectiveness increases at distances far from the Territorial Sea (Fig. 6, bottom centre).

The estimated effectiveness of SRS exhibited a complicated relationship with the covariates considered. In contrast to pilotage and VTS, the estimated effectiveness of SRS increased at distances far from the Territorial Sea baseline (Fig. 6, top right). Similarly to VTS, SRS effectiveness increased with axial alignment but at the scale of the scenario location rather than the surrounding region (Table 1). SRS effectiveness also decreased with the number of nautical miles travelled in the scenario location. In contrast to VTS, SRS may be made mandatory subject to SOLAS criteria and guidelines (Chapter 5, Regulation 19). However, without VTS, compliance to mandatory SRS measures is comparatively less monitored.

6. Discussion

An elicitation method may be judged by at least three criteria [*sensu* 20]: coherence, calibration and practicality. An elicited subjective probability assessment that respects the rules of probability calculus is said to be coherent. Calibration asks the question of how the elicited prior may perform when compared with new evidence. Practicality considers if the method provides for the participation of domain experts and the timely capture of their opinion.

Subjective probability distributions are coherent provided that such assessments satisfy the rules of probability calculus [2]. Cognitive biases and mental heuristics can lead unwary experts to provide incoherent assessments, for which the best remedy is proper education and careful facilitation [35]. Providing feedback and overfitting distributions during a facilitation helps experts review and check that their subjective probability statements for an unknown, but preferably observable, quantity are coherent and satisfactory to the expert [6]. The foregoing method for the elicitation of subjective probability distributions is based on the predictive approach for the unknown $\theta_i = \mathbb{E}[y_i]$, with observation model $p(y_i) = p(y_i|x_i, \beta, \xi)$ (Sections 2 and 3). Section 3 outlines a method that educates experts, provides feedback and overfits subjective probability distributions during the elicitation process. The problem of coherence across elicitation scenarios is difficult (Section 4.4) and discussion postponed to Section 6.1 below.

A fully coherent subjective probability distribution can be compared against new evidence, which leads to the question of calibration.

Table 1
Estimates from the best fitting model: median estimates and 90% central credible intervals. Covariates are preceded by *P, V, S* according to the existence of an interaction with the RCOs of pilotage, VTS or SRS, respectively. The covariates are *distTSB*, the log-transformed distance from the centre of the scenario location to the Territorial Sea baseline in nautical miles; nm, nautical miles travelled; *Ra*, axial resultant length in the scenario location; and *Ra21*, axial resultant length in the scenario neighbourhood. The vessel traffic covariates summarise Class A SOLAS vessels of 24 m length or more from AIS data.

	$\beta_{0.05}$	$\beta_{0.50}$	$\beta_{0.95}$
$P \times distTSB$	0.028	0.109	0.189
$V \times distTSB$	0.231	0.334	0.437
$V \times Ra21$	−2.112	−1.460	−0.808
$S \times nm$	0.035	0.064	0.093
$S \times distTSB$	−0.263	−0.165	−0.066
$S \times Ra$	−1.096	−0.599	−0.102
$P \times V$	−1.274	−0.797	−0.321
P	−1.779	−1.490	−1.201
V	−1.054	−0.764	−0.475
S	−0.685	−0.511	−0.336

Assume new evidence y becomes available after an elicitation session. It could be proposed to assess an elicited prior's predicted fractiles for y (such as median or quartiles, Section 3) against the observed fractiles for the empirical distribution of $p(y)$. A perfect expert is then one that conditions appropriately on all relevant information and translates opinion into the correct probability statements when predicting y. The perfect expert will be coherent and well-calibrated. An elicitation method however must accommodate the opinion of experts conditioned on current imperfect and incomplete information, not perfect opinion [20].

Education and feedback provide the best way to improve calibration when eliciting subjective probability distributions [50]. Experts may understandably adjust their opinion given new information, which is desirable in a scientific process; similarly, the best way to evaluate a given prior is through the collection of new empirical information such as by experimentation or monitoring [4]. For a given choice of model, M_j, the prior estimate $p(\beta|M_j)$ can be updated with future information using the Bayesian update $p(\beta|y_1, y_2, \ldots, y_K, M_j) \propto p(y_K|y_1, \ldots, y_{K-1}, \beta)p(y_1|\beta)p(\beta|M_j)$, for future observations y_k, $k = 1, \ldots, K$ that in principle can be obtained either empirically or via further elicitations. This allows coherent updating of the subjective probability estimate for β given new information ("coherence is all", D. V. Lindley [51]) and evaluation of expert opinion against the new relevant information.

The proposed approach was motivated by the practical challenge of eliciting predictive subjective probability distributions under model structure uncertainty. A key aspect is that the elicited subjective probability distribution are conditioned on scenarios that are each considered independently (Section 3). This approach follows advice developed for eliciting unknown events or parameters [6]. These elicited distributions are then used as data to form a prior for the unknown parameters β (Section 4), but the expert is insulated from this step for pragmatic reasons (Section 4.4). At this point an analyst takes on an unavoidable participatory role in the prior construction to ensure an appropriate representation of model structure and the uncertainty therein (Section 6.1).

6.1. Model structure uncertainty and elicitation

If one was facilitating infallibly perfect experts for a single agreed upon model, such that $J = 1$, then there would be no reason to assess $n > p$ elicitation scenarios. [25] focuses on the minimum $n = p$ case given the difficult and challenging nature of eliciting subjective uncertainty, or even $n < p$, which then requires a partial prior specification on the part of the analyst. To help alleviate workload, some approaches (e.g.,[22,24]) prompt experts with suggested predictions conditioned on a single known model structure. Although prompts run the risk of anchoring experts, it can be argued that a single model that restricts experts to considering only a few covariates limits the "dilution effect", where the perceived relationship between the predicted outcome and key covariates is obscured by extraneous covariates [6]. Certainly this approach would commit the elicited data to a single model structure.

However, there are cases where the model structure (i.e., covariates) is uncertain; elucidating model structure might in fact be the main goal of an elicitation session. The choice of model M_j is then a model selection or comparison problem. Model selection could take place prior to the elicitation, but there is a case to postpone selection or comparison until after the elicitation or even until the consideration of empirical data. For example, in Section 5, experts assessed key covariates before elicitation. Nevertheless, uncertainty remained in how the exact form of these covariates should be represented in the model, which might vary by RCO, interactions or other considerations. In fact, the approach of Section 5 presented key information, such as geographic location displayed on nautical charts, that entailed the consideration of a multitude of relevant factors, all of which could not possibly be addressed in a single model given the finite resources available for elicitation. This is itself a motivation for the assessment of subjective probability distributions that accommodate uncertainty for the specific scenarios in Section 3.

If the model structure itself is uncertain, then the facilitator will be forced to address the minimum number of scenarios to parametrise a proper prior for the biggest model. This condition is ensured by setting $n = p_{\max} = \max p_j$ assuming that all design matrices have full column rank. If $J > 1$ then a choice of model where $p_j < n = p_{\max}$ results in $n - p_j$ "extra" assessments allocated to $p(\beta|M_j)$ by the elicitation. The uncertainty in the prediction $X\beta$, as assessed by the trace of the prediction covariance, is then reduced by at least a factor of p_j/n determined by the size of model M_j (Appendix B).

Relative to a fully saturated model (where $p_j = n$), models of greater parsimony will have less uncertainty in the unknown parameters β. For these parsimonious models, there will be a greater weighting on the prior when assimilating an empirical data set into a Bayesian update. This effect is a feature of decreased uncertainty in β that follows naturally from Eq. (3), and could be viewed as the reward for choosing a parsimonious model. Although the above discussion was motivated by model structure uncertainty, there may be other reasons for conducting $n > p$ elicitation scenarios for a single model ($J = 1$). It seems sensible to guard against imperfection in the choice of model, experts and facilitator by allowing for such extra assessments, analogous to suggestions of overfitting when eliciting subjective probability distributions [6] and as adopted by the conditional elicitations of Section 3. This also has the effect of decreasing uncertainty in β conditional on a parsimonious model.

The way in which model structure uncertainty is assessed and represented can and arguably should change over the course of a Bayesian updating sequence. At the stage following the expert elicitation, a single model or suite of models may be selected for the purpose of prediction. BIC (Eq. (6)) for example may be used to assess model selection and minimise excessive complexity to improve predictions conditional on the elicited data. Now assume that new data y become available following an elicitation. Generally, a larger information set may support models of greater complexity. A wider suite of models may then be evaluated on the basis of y and the priors $p(\beta|M_j)$, $j = 1, \ldots, J'$, $J \leq J'$, which are proper if $p_j \leq n$ and $X|M_j$ is full column rank. These analyses can therefore motivate Bayesian model selection or comparison methods [29], such as Bayes factors or Bayesian model averaging [42], in addition to contributing to the posterior distributions $p(\beta|y, M_j)$. The use and selection of various model structures depends on the application and the information available.

7. Conclusions

A pragmatic approach to prior elicitation is proposed for eliciting parameters and model structure for Bayesian generalised linear models. The method was applied to an elicitation of Risk Control Option effectiveness for reducing the risk of ship collisions in Australia's Territorial Sea and Exclusive Economic Zone. The analysis suggests a synergistic interaction between the Vessel Traffic Services and pilotage RCOs. Predictive relationships with distance from the Territorial Sea Baseline and measure of shipping traffic patterns were documented. The proposed approach supports the coherent assimilation of new information and ultimately evidence based decision making.

Acknowledgements

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Appendix A

From Eqs. (1) and (2), the probability distribution of η conditional on the elicited data is given by,

$$\begin{aligned}\prod_{i=1}^{n} p(\eta_i|m_i,\upsilon_i) &= \prod_{i=1}^{n} p(x_i\beta|m_i,\upsilon_i)\\ &\propto \exp\left\{-\frac{1}{2}\sum_{i=1}^{n}\upsilon_i^{-1}(x_i\beta-m_i)^2\right\}\\ &= \exp\left\{-\frac{1}{2}\mathrm{Tr}\left[V^{-1}(X\beta-m)(X\beta-m)^{\top}\right]\right\}\\ &= \exp\left\{-\frac{1}{2}(X\beta-m)^{\top}V^{-1}(X\beta-m)\right\}. \qquad (A.1)\end{aligned}$$

which is proportional to the exponential of a quadratic form in β.

The distribution for the unknown β conditional on m and V will then be multivariate normal. From Eq. (A.1) and the expressions for μ (Eq. (4)) and Σ (Eq. (5)),

$$\begin{aligned}p(\beta|m,V) &\propto \exp\left\{-\frac{1}{2}\left[\beta^{\top}X^{\top}V^{-1}X\beta-2m^{\top}V^{-1}X\beta\right]\right\}\\ &= \exp\left\{-\frac{1}{2}(\beta-\mu)^{\top}\Sigma^{-1}(\beta-\mu)\right.\\ &\quad\left.+\frac{1}{2}X^{\top}V^{-1}m\left(X^{\top}V^{-1}X\right)^{-1}m^{\top}V^{-1}X\right\},\end{aligned}$$

which is proportional to the multivariate normal distribution, $N(\beta|\mu,\Sigma)$.

Appendix B

Denote the oblique projection into the column space of a $n\times p$ full column rank design matrix X by $H=X(X^{\top}V^{-1}X)^{-1}X^{\top}V^{-1}$. The matrix H is idempotent $H=HH$ with rank $\mathrm{Tr}[H]=p$. Then the prediction covariance is given by, $\mathrm{Cov}(X\beta)=HVH^{\top}=HV$. Define $h_i=H_{ii}$. The vector h can be shown nonnegative. Define $L=V^{-1/2}HV^{1/2}$. L is a symmetric idempotent matrix and also $L_{ii}=h_i$ for all i. By the properties of symmetric idempotent matrices, $L_{ii}=\sum_j^n L_{ij}^2=L_{ii}^2+\sum_{j\neq i}^n L_{ij}^2$ by which it is seen that $0\le L_{ii}\le 1$ and the bounds $0\le h_i\le 1$ are established for all i. Given nonnegative h and $\upsilon=[V_{11},\ldots,V_{nn}]^{\top}$,

$$\sum_{i=1}^{n} h_i\upsilon_i \le \|h\|_2\|\upsilon\|_2 \qquad (B.1)$$

$$\mathrm{Tr}[HV] \le \|h\|_1\|\upsilon\|_1 \qquad (B.2)$$

$$\mathrm{Tr}[HV] \le \frac{p}{n}\mathrm{Tr}[V]. \qquad (B.3)$$

Eq. (B.1) is the Cauchy–Schwarz inequality with the 2-norm $\|z\|_2=\left(\sum_{i=1}^n z_i^2\right)^{1/2}$. The right hand side of Eq. (B.2) follows from the norm inequality $\|z\|_2\le\|z\|_1=\sum_{i=1}^n|z_i|$. The left hand side of (B.2) follows from the definition of the matrix trace function and noting that V is diagonal so that $\sum_{i=1}^n h_i\upsilon_i=\mathrm{Tr}[HV]$. Lastly, $\|\upsilon\|_1=\mathrm{Tr}[V]$ and $\|h\|_1=\mathrm{Tr}[H]=p$. Eq. (B.3) follows by multiplying the constant $1/n$ with HV, h and υ in Eq. (B.2), then multiplying both sides of the inequality by n.

From Eq. (B.3), the average of the prediction variances $\mathrm{Cov}(X\beta)_{ii}=h_i\upsilon_i$ $(i=1,\ldots,n)$ is decreased by at least a factor of p/n relative to the average of the elicited variances $(\upsilon_i,\ i=1,\ldots,n)$. If V is homoscedastic with $\upsilon_1=\ldots=\upsilon_n=\kappa^2$ then the relationship (B.3) is an equality and the reduction factor is exactly p/n. A balanced design has $h_i=p/n$ for all i if V is homoscedastic (e.g.,[52]). For homoscedastic V and a balanced design, then every prediction variance is reduced by this same factor, that is, $h_i\upsilon_i=(p/n)\kappa^2$ for all i.

Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ress.2017.06.011.

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